TECHNICAL DOCUMENTATION & SOURCE CODE

Event Force Management System

Complete Database Schema & Core PHP/MySQL Implementation Modules

1. DATABASE SCHEMA SPECIFICATION (MYSQL)

The relational database `event_force_db` stores administrative accounts, volunteer details, scheduled events, shift allocations, and operational tasks.

database_schema.sql

```
CREATE DATABASE IF NOT EXISTS event_force_db CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci; USE event_force_db; -- 1. Admin Accounts CREATE TABLE admin ( admin_id
INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT NULL UNIQUE, password_hash
VARCHAR(255) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ) ENGINE=InnoDB; -- 2.
Staff Profiles CREATE TABLE staff ( staff_id INT AUTO_INCREMENT PRIMARY KEY, staff_name
VARCHAR(100) NOT NULL, phone VARCHAR(20) NOT NULL, email VARCHAR(100) NULL, role VARCHAR(50)
NOT NULL, status VARCHAR(20) DEFAULT 'Available' ) ENGINE=InnoDB; -- 3. Event Details CREATE
TABLE events ( event_id INT AUTO_INCREMENT PRIMARY KEY, event_name VARCHAR(100) NOT NULL,
event_date DATE NOT NULL, venue VARCHAR(150) NOT NULL, description TEXT NULL, status
VARCHAR(30) DEFAULT 'Planned' ) ENGINE=InnoDB; -- 4. Workforce Allocations CREATE TABLE
allocations ( allocation_id INT AUTO_INCREMENT PRIMARY KEY, event_id INT NOT NULL, staff_id
INT NOT NULL, assigned_role VARCHAR(50) NOT NULL, FOREIGN KEY (event_id) REFERENCES
events(event_id) ON DELETE CASCADE, FOREIGN KEY (staff_id) REFERENCES staff(staff_id) ON
DELETE CASCADE ) ENGINE=InnoDB; -- 5. Task Details CREATE TABLE tasks ( task_id INT
AUTO_INCREMENT PRIMARY KEY, allocation_id INT NOT NULL, task_description TEXT NOT NULL,
status VARCHAR(30) DEFAULT 'Pending', FOREIGN KEY (allocation_id) REFERENCES
allocations(allocation_id) ON DELETE CASCADE ) ENGINE=InnoDB;
```

2. DATABASE CONNECTION HANDLER

Establishes a connection to the MySQL database engine with UTF-8 character encoding.

db_config.php

```
<?php $servername = "localhost"; $username = "root"; $password = ""; $dbname =
"event_force_db"; // Establish Database Connection $conn = new mysqli($servername,
$username, $password, $dbname); // Check Connection Status if ($conn->connect_error) {
die("Database Connection Failed: " . $conn->connect_error); } // Set Encoding $conn-
>set_charset("utf8mb4"); ?>
```